# Use Case

## Autonomous Driving

- https://www.youtube.com/watch?v=vfirAm1ITMo
- Enabler for the future of driving
- Autonomous routing, obstacle detection, parking, …

[https://www.forbes.com/sites/enriquedans/2021/06/11/the-future-of-autonomous-vehicles-product-orservice/?sh=5ed0a2f5892f, March 28, 2023]

# Use Case

## Autonomous Driving

- Develop an autonomous vehicle that can drive autonomously on a given track
  - Based on line detection
- Being able to detect obstacles
  - Remove and driving around based on identifying the colour.
- Optimizing the speed
- Different routings
  - Driving in oval
  - Drive an eight
- Parking
- … and even more

# Prerequisite

- Create a team git
- Add all team members
- Add all lectures!
- Upload continuously your results to git
  - These includes the responsibilities
  - (Pre-) final version are uploaded within of the specified deadlines
- Divide the overall task into separate parts for each team-member in the following way, like:

| | | | Name1 | | Name2 | | Name… |
|---|---|---|---|---|---|---|---|
| # | Task | Short summary | Todo (incl. Deadline) | Done (incl. Finishing date | Todo | Done | … |
| 1 | Task1 | | | | | | |
| 2 | Task2 | | | | | | |
| … | Task… | | | | | | |

# Systems Engineering

## Task 1 & 2

HOCHSCHULE HAMM-LIPPSTADT

Task 1

- Develop a first system engineering model based on the Systems Engineering lecture
  - This includes all parts of the analysis
  - Outcomes are SysML Diagrams
    - Focus on requirements, including Requirements Diagrams, Use Case Diagrams, …

Task 2

- Refine your system engineering model and develop a first prototype
  - Being able to follow a line and detect obstacles
  - First version is simulated in tinkercad
  - Prerequisite for task 3 – using prototyping HW
  - System specification with SysML (pre final version)
  - Deadline: 24.05.26 eob

Task 3

- Milestone 1: Your vehicle should able to follow the lane(15.06.2026)
  - MileStone 2: Your vehicle should able to detect the obstacles (06.07.2026)

# Relevant criteria

- Quality of solution
  - Originality
  - Completeness
  - Integrity
- Usage of methods and techniques
  - Usage of process specific tools like github, trello, …
  - SysML/UML Diagrams like
    - Requirements, Use Cases, Scenarios, Constraints, Block-Diagrams, …